



SOMAIYA
VIDYAVIHAR UNIVERSITY

Semester: July 2023 – December 2023		
Maximum Marks: 30	Examination: In-Semester Examination	Duration : 1hr 15 min
Programme code: 01	Class: TY	Semester: V (SVU 2020)
Programme: B. Tech Computer Engineering		
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: COMP
Course Code: 116U01E511	Name of the Course: Computer Graphics	
Instructions: 1) Draw neat diagram where ever applicable 2) All Questions are compulsory		
3) Assume suitable data wherever necessary		

Question No.		Max. Marks	CO Mapped	BT Level
Q1	Attempt any TWO (2) A. In a 512 x 512 raster on a monochrome display with an average access rate of 200 nanoseconds per pixel, what is the refresh rate? B. Write a short note on GLUT C. Write a short note on homogeneous coordinates.	5 marks each	CO1 CO2	EV UN UN
Q2	Perform a 45° rotation of triangle A(0,0), B(1,1), C(2, 5) a. about origin b. about p(-1,-1)	10 marks	CO2	EV
Q3	Attempt any TWO (2) A. Explain aliasing and anti-aliasing effect in computer graphics. B. What is the purpose of the viewport transformation matrix, and how does it map normalized device coordinates to screen coordinates? C. Find the workstation transformation that maps the normalized device screen onto a physical device whose x extent 0 to 199 and y extent is 0 to 639 where the origin is located at the (a) lower left corner and (b) upper left corner if the device.	5 marks each	CO2 CO3	RE UN EV